Modelling Particle Production in Analog (3+1) Inflationary Cosmologies Using High-Performance Computing

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Introduction - Analog Universes

1. Inflation created particles

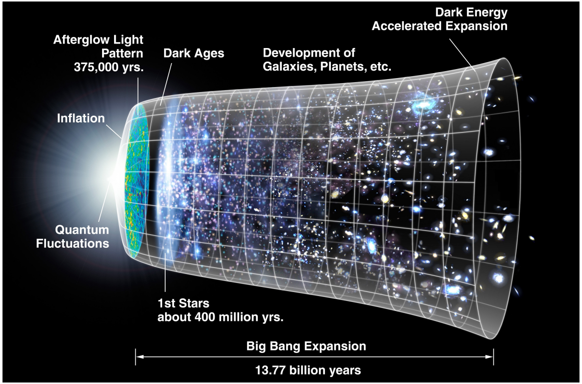

Figure: History of the universe highlighting period of rapid expansion (Source)

Introduction - Analog Universes

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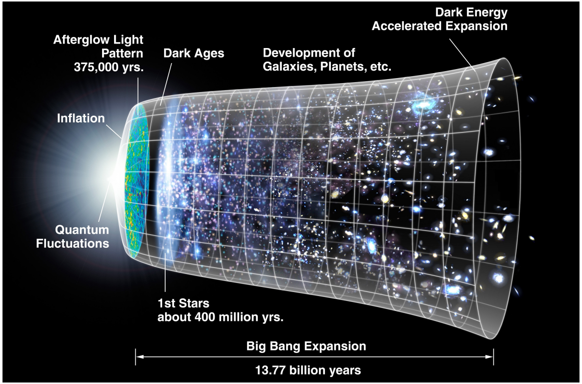
2. Cannot directly observe these particles



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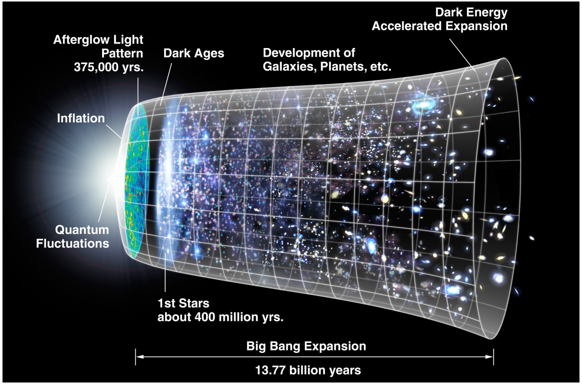

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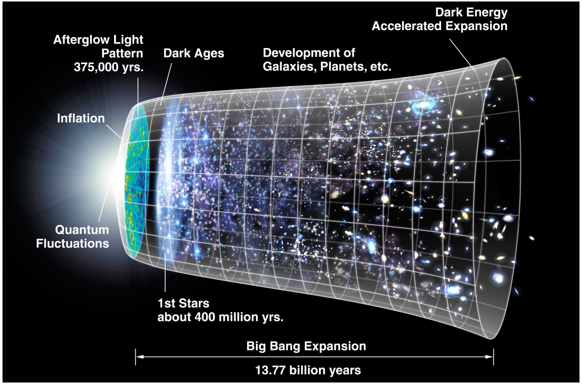
4. Can make gas in the lab $\implies$ can *effectively* observe inflation particles



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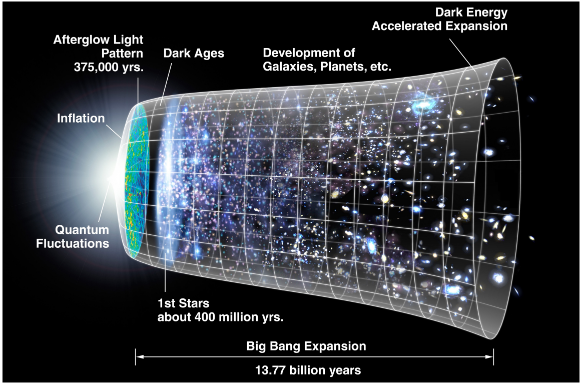
4. Can make gas in the lab $\implies$ can *effectively* observe inflation particles
5. Lab based system = "analog universe"



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Introduction - The Story So Far

- Many investigations into analog universes

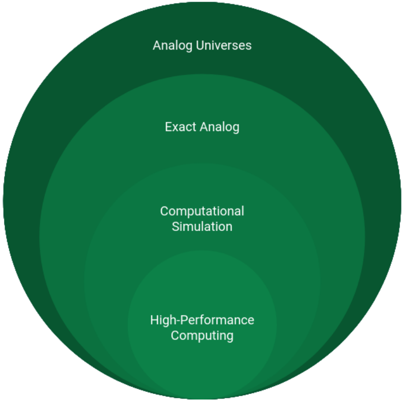

Figure: Subsets of increasingly specific literature

Introduction - The Story So Far

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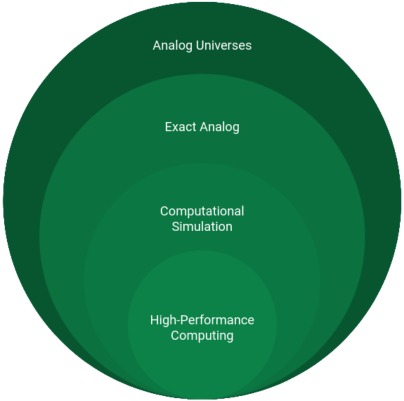
- Few into exact analogs



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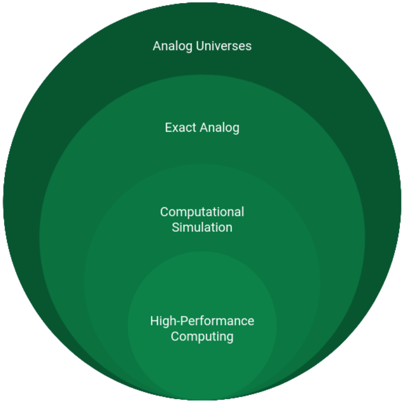

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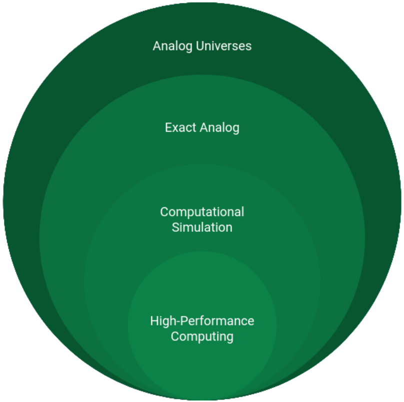
Introduction - The Story So Far

- Many investigations into analog universes
- Few into exact analogs
- Fewer into computational simulation
- Even fewer using high-performance computing



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Methods - The Field Equation

- One equation describes both ultra-cold quantum gases and inflation (a) particles

The Field Equation

$$\partial_t^2 \widetilde{\theta}_{\mathbf{k}} - \frac{\dot{a}}{a} \partial_t \widetilde{\theta}_{\mathbf{k}} - a c_0^2 k^2 \widetilde{\theta}_{\mathbf{k}} = 0$$

$$\widetilde{\theta}_{\mathbf{k}}(0) = \widetilde{\theta}_{\mathbf{k}0} \qquad \partial_t \widetilde{\theta}_{\mathbf{k}}(0) = -U_0 n_{\mathbf{k}0} / \hbar$$

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Methods - Symplectic Integrators

- Minimize energy error

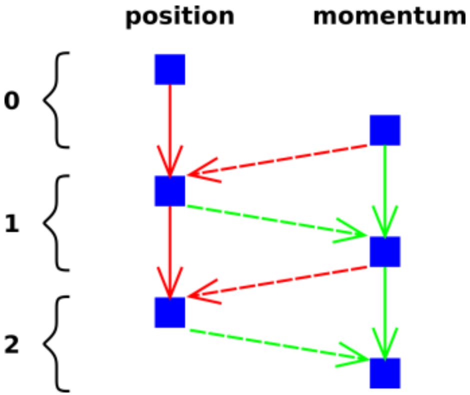

Figure: Symplectic integrator example: Velocity Verlet (Source)

Methods - Symplectic Integrators

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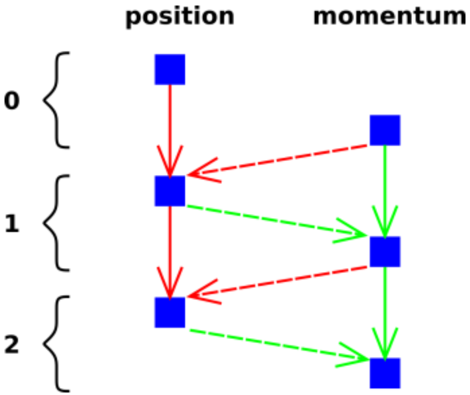

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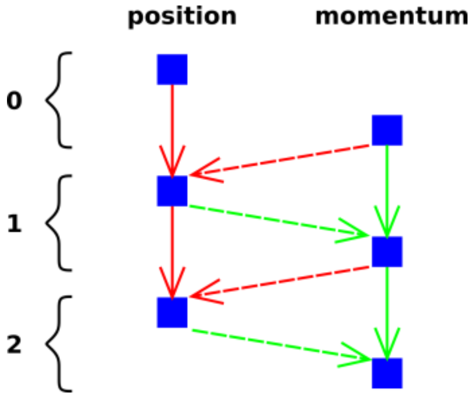

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Methods - Symplectic Integrators

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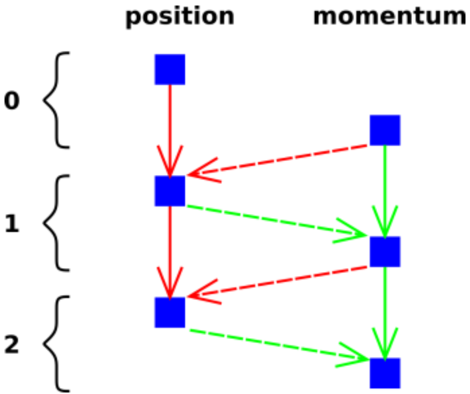
- Use new momentum to calculate position
- More complex but more stable



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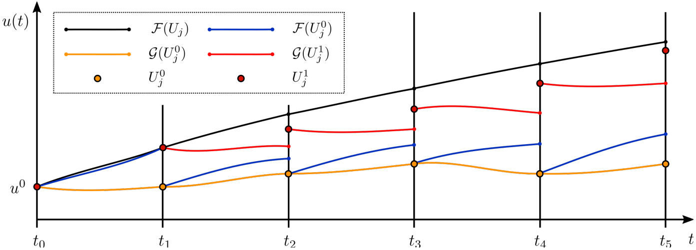

Figure: Illustration of the first iteration in Parareal (Source)

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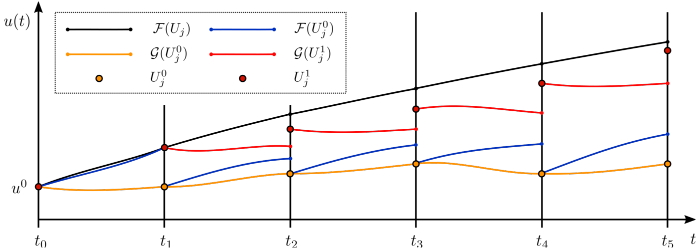

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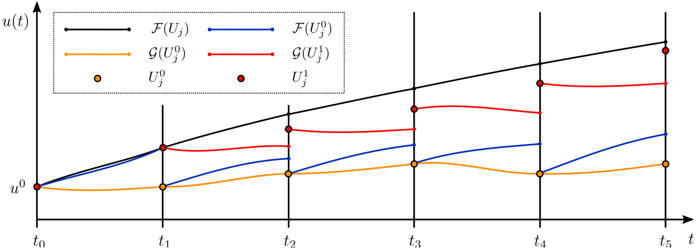

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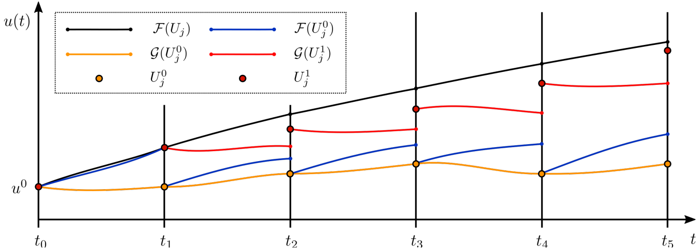

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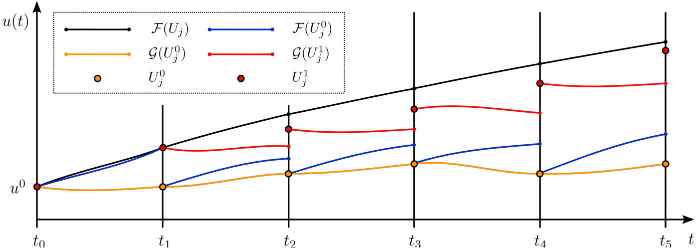
- Loop



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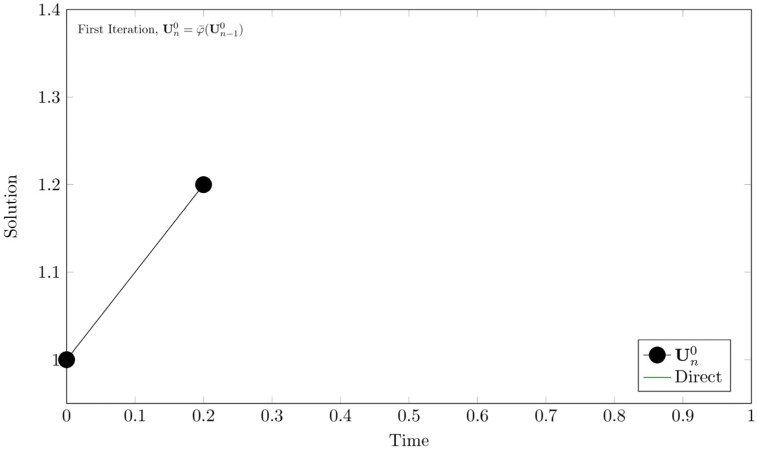

Figure: The *Parareal* approach to solving initial value problems in parallel.

Methods - High-Performance Computing

- Equation for each $k < k_c$

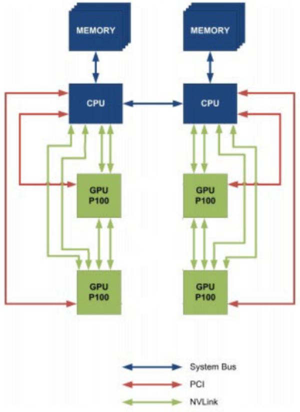

Figure: CPU to GPU and GPU to GPU interconnect using NVLink

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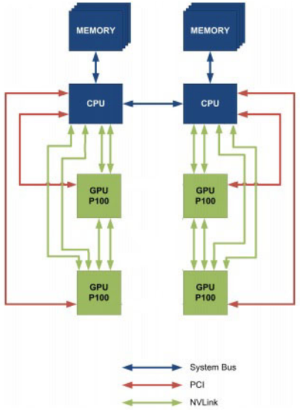

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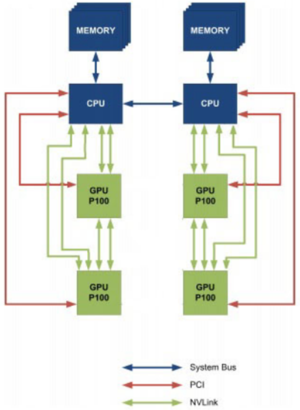

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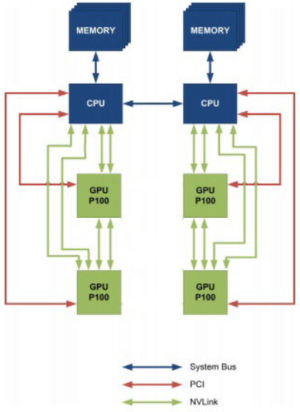
Methods - High-Performance Computing

- Equation for each $k < k_c$
- Many equations to solve
- Solve at the same time with multiple GPUs
- Use CWU's "Turing" supercomputer
- 12 NVIDA P100 GPUs
  - Each 83% performance of modern RTX 3060



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- Simple expansion: $a(t) = e^{-t/t_s}$

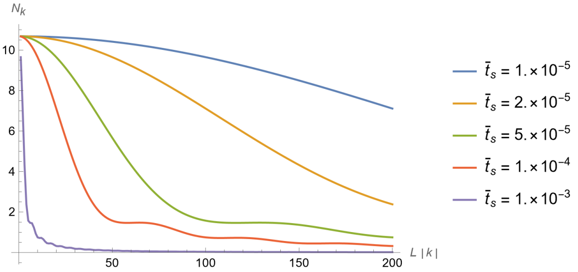

Figure: The momentum distribution when the universe stops expanding for multiple effective rates of expansion

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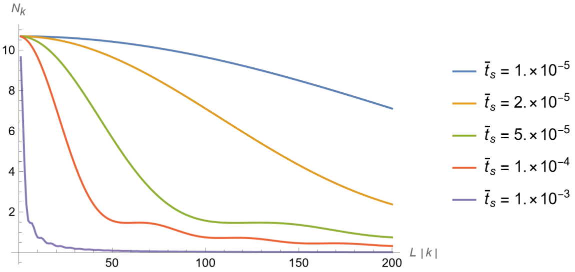

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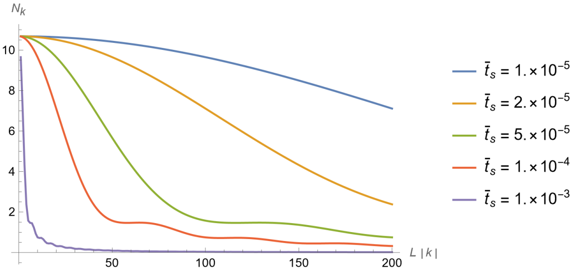

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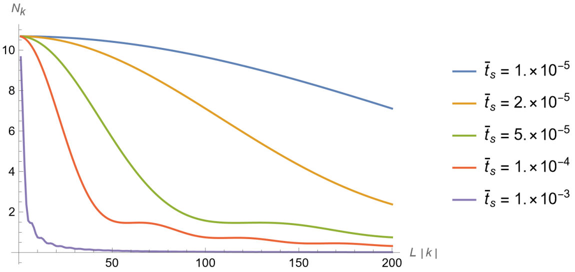

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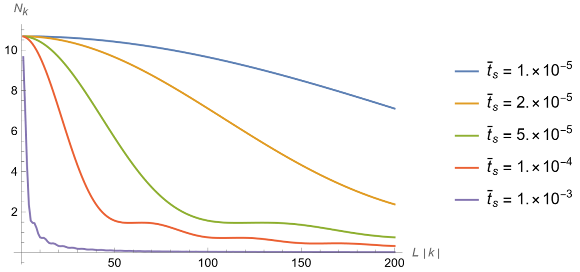

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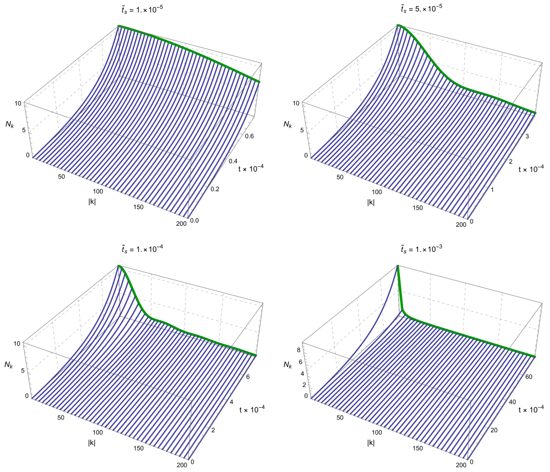

Figure: Momentum distribution throughout the expansion

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- Use high-performance techniques and hardware for reasonable execution time
- Expect the distribution of momenta to depend on the length of the expansion
- Expect the population to be high for low k and drop off quickly

Acknowledgements

- Dr. Andy Piacsek (CWU, Physics)
- Dr. Szilard VAJDA (CWU, Computer Science)
- Dr. Micheal Braunstein (CWU, Physics)
- Dr. Brandon Peden (WWU, Physics & Astronomy)

Thank you & Code source

Thank you for your attention.

The code for this project is available at:

- GitHub (QR code): `github.com/nondairyneutrino/thesis`
- My website: `nathanwchapman.com`